

A Dynamic Ankle Orthosis Reduces Tibial Compressive Force and Increases Ankle Motion Compared With a Walking Boot

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ABSTRACT

JOHNSON, P. A., M. R. PAQUETTE, and D. J. DIANGELO. A Dynamic Ankle Orthosis Reduces Tibial Compressive Force and Increases Ankle Motion Compared With a Walking Boot. *Med. Sci. Sports Exerc.*, Vol. 55, No. 11, pp. 2075–2082, 2023. **Purpose:** Tibial bone stress injuries are a common overuse injury among runners and military cadets. Current treatment involves wearing an orthopedic walking boot for 3 to 12 wk, which limits ankle motion and leads to lower limb muscle atrophy. A dynamic ankle orthosis (DAO) was designed to provide a distractive force that offloads in-shoe vertical force and retains sagittal ankle motion during walking. It remains unclear how tibial compressive force is altered by the DAO. This study compared tibial compressive force and ankle motion during walking between the DAO and an orthopedic walking boot. **Methods:** Twenty young adults walked on an instrumented treadmill at 1.0 m·s⁻¹ in two brace conditions: DAO and walking boot. Three-dimensional kinematic, ground reaction forces, and in-shoe vertical force data were collected to calculate peak tibial compressive force. Paired *t*-tests and Cohen's *d* effect sizes were used to assess mean differences between conditions. **Results:** Peak tibial compressive force ($P = 0.023$; $d = 0.5$) and Achilles tendon force ($P = 0.017$; $d = 0.5$) were moderately lower in the DAO compared with the walking boot. Sagittal ankle excursion was 54.9% greater in the DAO compared with the walking boot ($P = 0.05$; $d = 3.1$). **Conclusions:** The findings from this study indicated that the DAO moderately reduced tibial compressive force and Achilles tendon force and allowed more sagittal ankle excursion during treadmill walking compared with an orthopedic walking boot. **Key Words:** DYNAMIC ANKLE ORTHOSIS, WALKING BOOT, TIBIAL COMPRESSIVE FORCE, ACHILLES TENDON LOADING, REHABILITATION

Running is a common form of exercise that has many health benefits, but unfortunately, overuse injury rates and incidences are high among distance runners. The repetitive nature of running at submaximal levels causes the mechanical properties of bone to degrade over time, leading to increased strain and the development of microfractures. This phenomenon is commonly referred to as fatigue (1–4). When provided with sufficient time to rest, bone can remodel and adapt to the loading environment (5). However, when the rate of bone damage outweighs the rate of repair, microcrack propagation can lead to stress fractures (1,6–8). Tibial bone stress injuries (TBSI) are a common overuse injury accounting for 21.9%–69% of stress fracture incidence among runners (9,10) and 24%–51.2% of stress fractures in military cadets (11–13). TBSI most commonly occur in the middle to distal third of the anterior tibia from rapid increases in physical activity or running exposure (14).

A period of reduced loading experienced at the injured site is often recommended after the initial diagnosis of a TBSI (15). Current standard of care consists of TBSI patients wearing a walking boot to immobilize the lower limb during the recovery period. The severity of the stress fracture will often dictate the period of time the TBSI patient will wear a boot. Even for low-risk stress fracture injuries that are generally treated with conservative methods compared with high-risk stress fracture injuries that require surgical treatment, patients are typically prescribed boot wear for 3–12 wk, depending on the level of severity of the injury, before returning to running activities (13,15–17). General rehabilitation methods for handling stress fractures are separated into two phases. Phase 1 (acute phase) requires a period of immobilization and reduced loading where rest, ice, compression, and elevation is needed (18). When the patient is pain-free, they can enter the second phase of rehab incorporating progressive loading (15,16,18). Although walking boots are the current standard of care, there is no knowledge if, or how, they affect loading mechanics on the tibia.

Most studies on TBSIs have focused on clinical outcomes (13,19,20) and do not examine the mechanical loading environment of bone when wearing a walking boot. Findings from studies focused on the biomechanics of walking boots found that they provide a reduction in forefoot pressure, but conflicting evidence suggest they moderately reduce or increase heel pressure (21–24). Walking boots also limit sagittal ankle motion

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causing reduced muscle activity, which could, and generally does, lead to muscle atrophy of the gastrocnemius, soleus, and peroneal muscles (25–28). Muscular atrophy contributes to prolonged rehabilitation time, which delays return to training or competition in athletes. Furthermore, Bailón-Plaza and Van Der Meulen (29) studied fracture healing of sheep tibia under different loading and timing of mechanical stimulations, and determined the axial loading stiffness over a 100-N load range divided by the average nodal displacement of the cortex. They observed that moderate mechanical loading provides a positive loading environment for quicker bone ossification and increased stiffness at the injured site compared with intact bone. By week 12, the stiffness of the stimulated bone group had reached 70%–94% of the intact bone compared with half of the unstimulated bone group reaching 70%–80% stiffness and the other half reaching 30%–33% stiffness of the intact bone. Furthermore, devices that allow appropriate mechanical loading of the tibia and minimize lower limb muscle atrophy during rehabilitation may help improve return to training or competition time in athletes or military personnel. Surprisingly, no studies have assessed the mechanical loading of the tibia in a walking boot, or in alternative rehabilitation devices, during daily walking activities.

A novel dynamic ankle orthosis (DAO), consisting of a calf sleeve, modified shoe, and two pneumatic cylinders (Fig. 1A), was recently designed to apply a distractive force to offload the foot and ankle while also retaining natural ankle motion during level walking (30). During treadmill walking when compared with a no-brace condition, the DAO was able to significantly reduce peak plantar pressure under toes 2–5 by 12% and under the hallux by 24%, whereas a double upright ankle foot orthosis increased peak plantar pressure under toes 2–5 and hallux by 14% and 6%, respectively (30). A second study assessing the effects of acute DAO wear on perceived function and underfoot forces in symptomatic patients who have experienced mechanical foot and ankle pain symptoms found that the DAO moderately reduced vertical in-shoe force compared with a no-brace condition during overground walking (31).

Although the DAO has been shown to reduce peak vertical in-shoe force while retaining natural ankle motion in healthy adults, its effectiveness to reduce the loading environment experienced by the tibia while walking remains unknown. The rationale behind the DAO is to reduce the loading environment to theoretically reduce the stress and therefore reduce the strain on the bone and improving healing. The purpose of this study was to determine the biomechanical effects of a DAO on tibial bone loading and sagittal ankle motion compared with a standard-of-care walking boot in healthy individuals during treadmill walking. We hypothesized that the DAO would reduce the tibial compressive force and allow greater sagittal ankle excursion compared with the walking boot.

METHODS

Participants. *A priori* power analysis using an effect size f of 0.32 (from estimated Achilles tendon force between a walking boot and the DAO in preliminary analyses), α set to 0.05, and power of 0.8 for two within-subject repeated measurements suggested that a minimum of 18 participants were necessary to obtain the statistical effect size and power to test our hypotheses. Therefore, 20 participants were recruited (10 men, 10 women; 23 ± 2.4 yr; 1.7 ± 0.01 m; 69.1 ± 11.4 kg) for this study. Inclusion criteria were participants between the ages of 18 and 40 yr having a body mass less than 90 kg (due to instrument weight limitations) and no lower limb or low back injuries in the past 12 months. Participants were excluded if they had any chronic disorders or orthopedic conditions that could influence walking mechanics (including lower back pain or injury), or if they were pregnant at the time of the study. Participants were informed of all procedures and potential risks, and signed a written consent form approved by the institutional review board before data collection.

Orthosis conditions. Two brace conditions were tested: the DAO (0.78 ± 0.12 kg) and a standard-of-care walking boot (Össur FormFit Walker Air; 0.5 ± 0.06 kg; Fig. 1B). The shoe used for the DAO condition (model 501; New Balance,

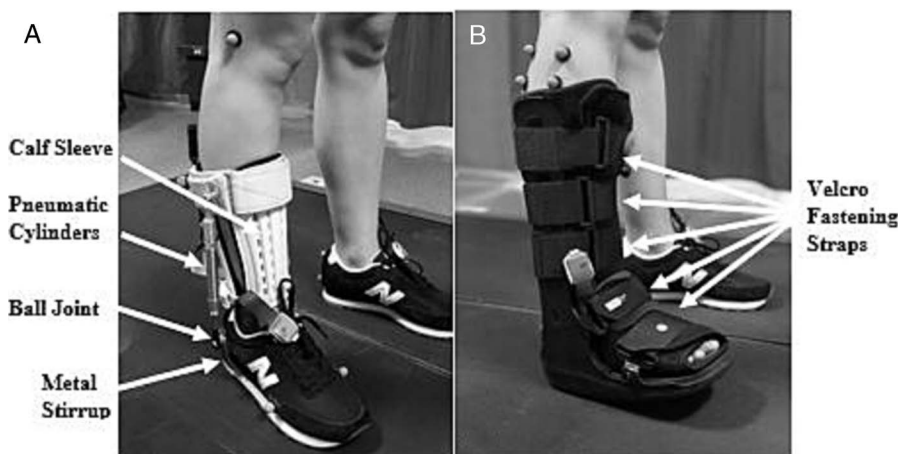


FIGURE 1—Brace conditions: DAO (A) and walking boot (B).

Boston, MA) was modified with a metal stirrup and plate embedded in the sole of the shoe, which provided connection to the upper orthosis components. Previous validation testing showed that the DAO provided comparable ranges of movement to the intact ankle joint allowing for at least 10° of dorsiflexion, 15° of plantarflexion, and at least 10° of inversion and eversion (30). In a previous study with the DAO, a targeted offloading of at least 10% of the participants' body weight (BW) was recommended by orthotists regarding the level of offloading necessary to achieve pain reduction (30). In this study, the DAO was activated to offloaded 10% BW based on the static protocol outlined hereinafter. The heel portion of the foam material of the walking boot was removed to secure reflective markers on the heel to track foot motion during testing.

Experimental protocol. The two brace conditions were randomized and worn on the participant's right limb. For the DAO condition, static offloading (10% BW target) was determined during standing by taking the percent difference in the insole force reading before and after pressurizing the pneumatic cylinders. The static protocol was performed five times per participant to ensure proper offloading. Static offloading of $12.6\% \pm 0.02\%$ was achieved from testing. Each brace condition was worn for 5 to 10 min during both standing and walking before data collection to provide a period of brace familiarization (28,30,32–34). Meyer et al. (32) and Zeni and Higginson (33) have shown this familiarization time to be sufficient for kinematic and kinetic variables, respectively.

Individual retroreflective markers were positioned on the first and fifth metatarsal heads, medial and lateral malleoli, and medial and lateral femoral epicondyles to define segment dimensions and joint coordinate systems, and to establish a relationship between joint centers and anatomical tracking markers. Clusters of tracking markers were placed on the right rearfoot and shank in different positions for each brace condition to track each segment. For the DAO condition, tracking markers were positioned on the posterior region of the shank on the DAO calf sleeve component. A heel cluster of three tracking markers rigidly attached to a thermoplastic plate was securely positioned on the posterior aspect of the shoe. For the walking boot condition, a cluster of four tracking markers were positioned at the posterior–lateral and proximal portion of the shank above the walking boot brace. An eight-camera three-dimensional (3D) motion capture software with Qualisys Track Manager software (240 Hz; Qualisys AB, Gothenburg, Sweden) captured 3D kinematic data, whereas a force instrumented split-belt treadmill (1200 Hz; Bertec, Inc, Columbus, OH) collected 3D ground reaction force (GRF) data and a wireless vertical force-sensing insole (100 Hz, loadsol; Novel, St. Paul, MN) was inserted into the braced shoe to measure in-shoe vertical reaction force (RF_z) during treadmill walking. Insole force data were calibrated using the manufacturer procedure before testing in each brace condition. After a 1-s standing calibration trial, anatomical markers were removed before treadmill walking. Participants walked on the 3D force treadmill at $1.0 \text{ m} \cdot \text{s}^{-1}$ for 5 min to become acclimated to each brace condition before

data collection. After walking on the force treadmill for an additional 2 min, 3D kinematic, kinetic, and insole vertical force data were captured for 30 s. At the start of each walking condition, participants took a forceful step with their right foot on the treadmill to identify an instant in time, detected from all three instruments, to synchronize all data. Participants were allowed to rest for at least 5 min, or until they felt well rested, between brace conditions to minimize fatigue effects. From the 30-s worth of data captured, nine sequential steps were used from each brace condition for data analysis. After each testing trial, participants rated their comfort level in each brace condition using a custom visual analog scale. Comfort was rated with 0, “very comfortable”; 2, “comfortable”; 4, “moderately comfortable”; 6, “moderately uncomfortable”; 8, “uncomfortable”; and 10, “extremely uncomfortable.”

Data analyses. 3D kinematic, insole force, and GRF data were processed and analyzed for nine sequential steps using Visual 3D software (C-Motion, Germantown, MD) and MATLAB (MathWorks, Natick, MA). The raw kinematic, insole force, and GRF data were interpolated to fill data gaps no greater than 10 frames and filtered using a Butterworth low-pass filter with cutoff frequencies of 6, 15, and 15 Hz, respectively. Right-hand rule with Cardan rotational sequence (x – y – z) was used for 3D angular kinematic computations, where x represents rotations about the mediolateral axis, y represents the rotation about the anteroposterior axis, and z represents the rotation about the vertical axis of the distal segment. Sagittal plane ankle kinematics were resolved in the shank coordinate system, and all variables were analyzed during the stance phase of gait. A vertical force threshold of 25 N was used to define heel contact and toe-off for both the insole force and GRF data. 3D kinematic, treadmill kinetic, and insole force data were used to calculate the moment acting about the ankle joint at time of peak plantarflexor moment. High validity and day-to-day reliability have been shown when comparing GRF variables from the loadsol compared with an instrumented treadmill (35).

The 3D kinematic and treadmill GRF data were time-synced with the time point of the insole RF_z data by the participants taking a forceful step with their right foot to synchronize kinematic and kinetic data to determine the net ankle moment at peak plantarflexor moment for inverse dynamic calculations (Fig. 2). Because the force treadmill did not account for any reduction in the vertical GRF component due to the offloading effect of the DAO, the vertical GRF vector was replaced with RF_z . It is also important to note that the information from the instrumented force treadmill and force-sensing insole provides different force vector data. The instrumented treadmill provided the resultant GRF, whereas the force-sensing insole provided the vertical force component only. An image of the participants' lateral right foot, from the lateral malleolus to Achilles tendon, was taken and uploaded to ImageJ (National Institutes of Health, University of Wisconsin). From this image, the anteroposterior distance between the lateral malleolus and the superficial surface of the Achilles tendon was determined to represent the sagittal plane Achilles tendon moment arm (d_{AT}) to the ankle joint. Variables used in the calculations were extracted in Visual 3D

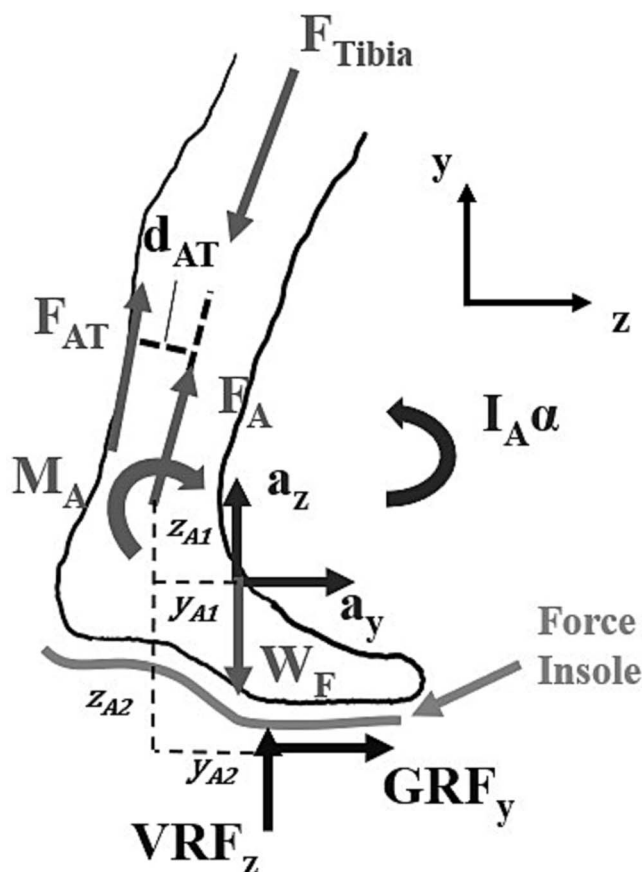


FIGURE 2—Free body diagram of tibial force calculation.

(excluding RF_z) and include the moment about the ankle, M_A ; moment of inertia about the ankle, I_A ; angular acceleration, α ; mass of the foot, M_F ; linear acceleration, a_y ; linear acceleration moment arm, z_{A1} ; vertical acceleration, a_z ; vertical acceleration moment arm, y_{A1} ; RF_z , RF_z moment arm, y_{A2} ; propulsive GRF (GRF_y), GRF_y moment arm, $z_{A1} + z_{A2}$; and gravity, g , as seen in equation 1 hereinafter.

$$M_A = I_A \alpha - [(m_F \times a_y) \times z_{A_1}] - [(m_F \times a_z) \times y_{A_1}] - (R F_z \times y_{A_2}) \quad [1]$$

$$- [GRF_y \times (z_{A_1} + z_{A_2})] + [(m_F \times g) \times y_{A_1}]$$

$$F_{\text{AT}} = \frac{M_{\text{A}}}{d_{\text{AT}}} \quad [2]$$

$$F_A = \frac{F_{z'}}{\cos(\beta)} \quad [3]$$

$$F_{\text{tibia}} = F_{\text{AT}} + F_{\text{A}} \quad [4]$$

Achilles tendon force, F_{AT} , represents the muscle forces acting on the tibia. F_z' represents the summation of the vertical forces acting on the tibia (i.e., equal and opposite to the summation of forces at the ankle, F_z) at angle β (angle of tibia at peak plantarflexor moment) and used to obtain the reaction force at the ankle, F_A , acting along the longitudinal axis of the tibia. The peak compressive force, F_{tibia} , acting on the tibia

was the addition of the ankle reaction force, F_A , and muscular forces, F_{AT} (36–38). Peak plantarflexor moment was chosen as the specific time of interest because it is the point in stance phase, at the start of push-off, where the calf muscles are most involved in assisting with propelling the body forward (39). The RF_z at the time of peak plantarflexor moment was extracted from the time series using a custom MATLAB script. Sagittal plane ankle excursion was defined by the period after heel contact to peak dorsiflexion after mid stance. Vertical impulse was determined as the area under the force time interval from the force insoles. Stance time and step length were measured during gait. The percent offloading provided by the DAO compared with the walking boot was determined by taking the difference between the peak force in the DAO, $Force_{DAO}$, and the peak force in the walking boot, $Force_{WB}$, divided by the peak force in the walking boot (equation 5).

$$\% \text{offloading} = \frac{\text{Force}_{\text{WB}} - \text{Force}_{\text{DAO}}}{\text{Force}_{\text{WB}}} \times 100 \quad [5]$$

Statistical analyses. A Shapiro–Wilk test was used to assess normality of the data per condition for each dependent variable. A paired *t*-test was used to determine brace effect for all dependent variables that passed normality. Cohen’s *d* effect size was determined to assess the effect size for differences between dependent variables (i.e., small: $d < 0.2$, medium: $0.2 \leq d < 0.8$, large: $d \geq 0.8$). Step length, cadence, and brace comfort score failed normality (i.e., nonparametric); therefore, a Wilcoxon sign rank test was used to compare means. A significance level of $P < 0.05$ was used for all tests.

RESULTS

Peak tibial compressive force (10.8%; Fig. 3) and Achilles tendon force (11.9%; Fig. 4) were lower in the DAO compared with the walking boot (Table 1). Ankle dorsiflexion excursion (Fig. 5) was reduced by 54.9% in the walking boot compared with the DAO (Table 1). Stance time was longer in the DAO compared with the walking boot (Table 1). Vertical impulse, step length, cadence, and brace comfort rating were not significantly different between brace conditions (Table 1).

DISCUSSION

The overall goal of this study was to compare the effects of a DAO on lower extremity loading biomechanics during treadmill walking with those of a standard-of-care walking boot in young healthy adults. Peak tibial compressive force was lower in the DAO compared with the walking boot. This finding supports our first hypothesis that the DAO moderately reduced the tibial compressive force and further confirms that the recommended DAO offload setting of 10% was sufficient to create a significant difference in the amount of tibial force reduction (i.e., 10.8%). Given the calculations for tibial compressive force, the lower estimated peak Achilles tendon force ($d = 0.51$) due to lower net peak plantarflexor moment explains the lower peak tibial compressive force ($d = 0.48$) in

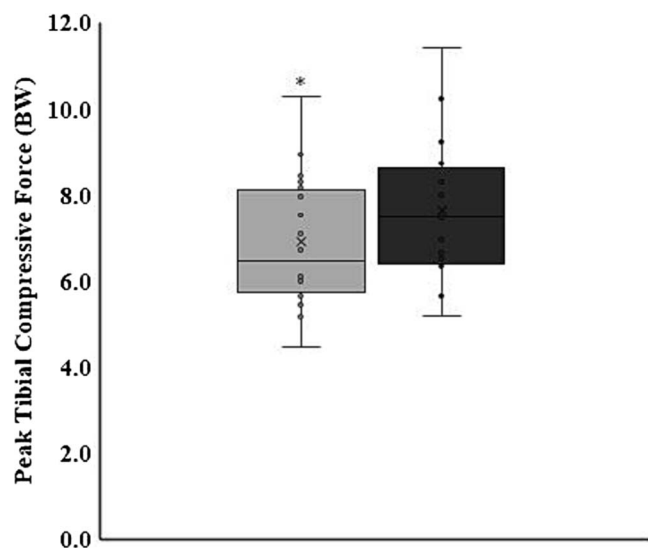


FIGURE 3—Peak tibial compressive force at peak plantarflexor moment during treadmill walking of the right limb for the DAO and walking boot conditions. *Significant difference compared with walking boot ($P < 0.05$).

the DAO. Furthermore, at the time of peak plantarflexor moment, there was no difference in the RF_z reported by the force insoles between the DAO and walking boot. However, the shorter moment arm of RF_z relative to the ankle joint explains the lower peak plantarflexor moment in the DAO compared with walking boot (Fig. 6). The shallow rocker sole contributed to less dorsiflexion throughout the stance phase, and rigid design of the walking boot was responsible for the longer moment arm of RF_z relative to the ankle joint at the time of peak plantarflexor moment. On the other hand, the DAO had a shorter forefoot rocker point (and moment arm) compared with the walking boot, and the unconstrained ankle joint of the DAO allowed the participant to raise their heel and pivot about the metatarsophalangeal joint in late stance, which reduced the moment arm of RF_z relative to the ankle joint. With extended DAO wear during the healing stages of a TBSI, bone union may occur at a faster rate allowing quicker return to sport. Thus, future studies should investigate the influence of DAO wear during a TBSI recovery phase to possibly improve bone healing compared with a standard-of-care walking boot.

Of additional clinical interest, the lower Achilles tendon force in the DAO could potentially benefit individuals recovering from Achilles injury. Current evidence on Achilles tendon rehabilitation suggests that progressive loading (i.e., lower to higher tensile forces) of the tendon is most effective (40). Thus, use of the DAO in early stages of Achilles tendon rehabilitation or possibly after Achilles tendon surgery to keep tendon loading lower may be of clinical interest. Furthermore, and in agreement with our second hypothesis, the DAO allowed greater sagittal ankle excursion compared with the walking boot from heel contact to peak dorsiflexion in late stance. The smaller ankle excursion observed in the walking boot (13°) may limit or reduce muscle activity and possibly contribute to muscle

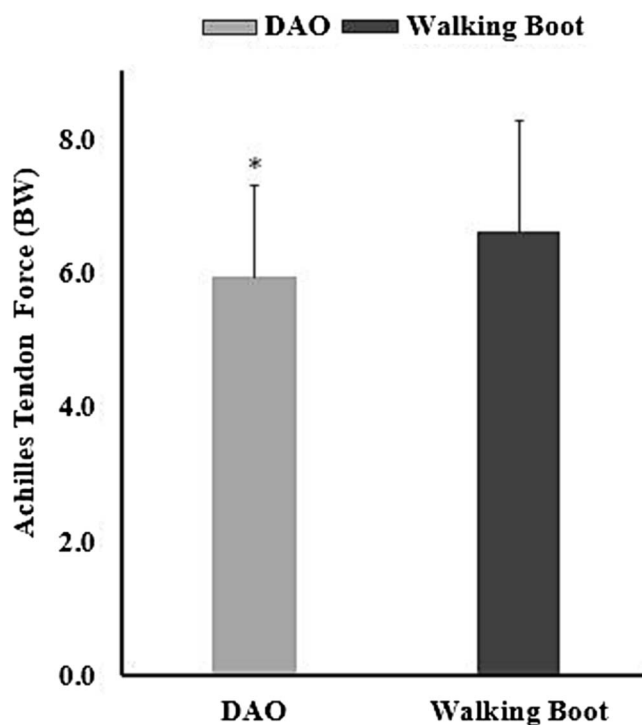


FIGURE 4—Achilles tendon force at peak plantarflexor moment during treadmill walking of the right limb for the DAO and walking boot conditions. *Significant difference compared with walking boot ($P < 0.05$).

atrophy in the gastrocnemius, soleus, and peroneal muscles (25–28). Nahm et al. (41) observed that participants experienced between 3° and 8° of sagittal tibiotalar motion while wearing the different walking boot conditions compared with a regular shoe. Zhang et al. (42) observed between 5° and 7° of sagittal ankle excursion in individuals wearing short-leg walking boots during walking. Although the DAO allowed more ankle excursion, to a degree similar to natural sagittal plane ankle motion (25° – 30°) (43,44) and may provide more involvement of the plantarflexor muscles, it remains unknown if muscle atrophy would occur during extended walking boot wear during the recovery phase of a post-TBSI. Thus, future studies should investigate the influence of DAO wear during a TBSI recovery phase to observe the amount of muscle atrophy experienced compared with a standard-of-care walking boot.

TABLE 1. Kinetic, kinematic, and spatiotemporal variables for each brace condition at prescribed treadmill walking speed (mean $\pm$ SD).

Variables	DAO	Walking Boot	P	Effect Size (d)
Peak tibial compressive force (BW)*	6.9 $\pm$ 1.5	7.6 $\pm$ 1.7	0.030	0.45
Achilles tendon force (BW)*	5.9 $\pm$ 1.4	6.6 $\pm$ 1.6	0.021	0.49
Dorsiflexion excursion ($^\circ$)*	29.2 $\pm$ 3.0	13.2 $\pm$ 4.1	<0.001	3.19
Vertical impulse (N-s)	420.2 $\pm$ 76.0	414.1 $\pm$ 73.1	0.075	0.34
Stance time (s)*	0.77 $\pm$ 0.04	0.74 $\pm$ 0.04	<0.001	1.07
Step length (m)	0.58 $\pm$ 0.05	0.57 $\pm$ 0.03	0.198	0.20
Cadence (steps per minute)	105 $\pm$ 11	105 $\pm$ 6	0.422	0.13
Brace comfort scores	3.9 $\pm$ 2.6	5.0 $\pm$ 2.7	0.108	0.29

Brace comfort scores: 0 = very comfortable, 10 = very uncomfortable. Statistical analyses for step length and cadence were performed using a Wilcoxon sign rank test.

*Brace effect ($P < 0.05$).

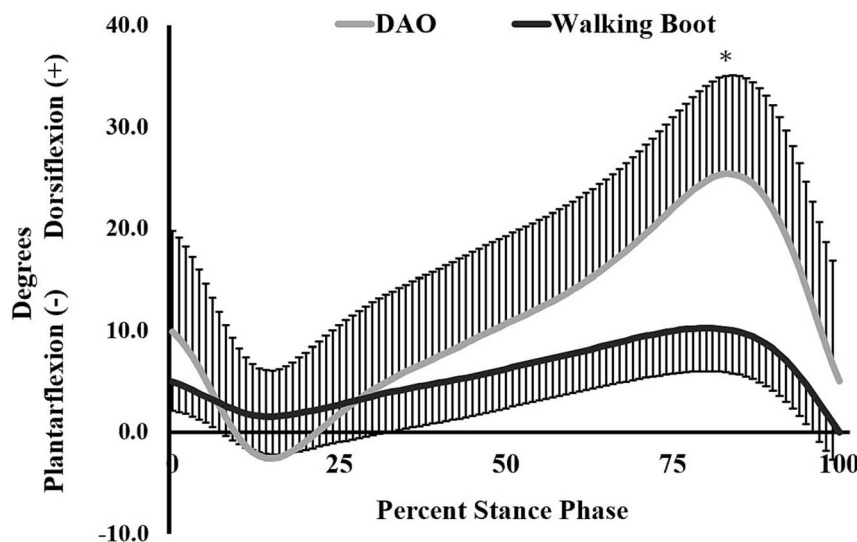


FIGURE 5—Sagittal plane ankle joint curves during treadmill walking of the right limb for the DAO and walking boot brace conditions. *Significant difference compared with walking boot.

The Achilles tendon loads determined in this study were comparable to other findings in the literature (45,46) (Fig. 7A). Hullfish et al. (45) observed Achilles tendon loads of 2 BW compared with our results of 5.8 BW and 6.6 BW in the DAO and walking boot, respectively. Hullfish et al. developed a simplified protocol to quantify the load at the Achilles tendon using Loadsol insole vertical force data and a load cell attached to the inline of the posterior strut of a walking boot but did not measure ankle excursion. Their assumption that the ankle remained fixed in the walking boot and did not allow for any ankle movement created an overconstrained system that negated any Achilles tendon involvement, which may have led to a lower estimation of the Achilles tendon loading. Our model used a motion capture system and found that the walking boot allowed some ankle joint movement (Fig. 4). Some discrepancies exist on where the moment arm length was measured from that could explain the larger differences in the Achilles tendon force values. Our moment arm of 3.1 cm was measure relative to the lateral malleoli, whereas Hullfish et al. reference other studies (citing 5.4 cm) (47,48) as measured from the medial malleoli or midpoint between the lateral and medial malleoli.

Figure 7B provides a comparison of our tibial force values with other biomechanical studies (36,37,46) for walking and running. In the walking study by Meardon et al. (46), the peak tibial bone loads for a group of recreational runners were 5000 N (6.36 BW) and 3800 N (6.65 BW) in men and women, respectively, and were very similar to our findings. Scott and Winter (36) developed a mathematical model to predict the loads at chronic injury sites during running. They observed an average peak tibial compressive force of 12.32 BW during stance phase. As one would expect, the peak tibia loads during running were greater than our measured values for walking.

The findings of this study are limited to acute brace wear because participants were only allowed 5 to 10 min to become acclimated to the brace condition. To understand the long-term

effects on bone health and/or healing from DAO use compared with a walking boot, a randomized control trial over multiple weeks is needed. Brace sizes were limited to three options for the DAO (small, medium, and large) and two options for the walking boot (medium and large). Any brace adjustments were limited to what was feasible with available brace sizes rather than creating patient-specific braces. The shoe worn on the participants contralateral limb was not adjusted to account for limb length discrepancies; however, this is reflective of how, in a clinical setting, patients will often wear their own shoe during the prescribed walking boot period. Therefore, our results are reflective of most clinical scenarios. Peak impact and push-off forces have been shown to differ between overground and treadmill walking when wearing the DAO (31). The differences in sampling rates between the

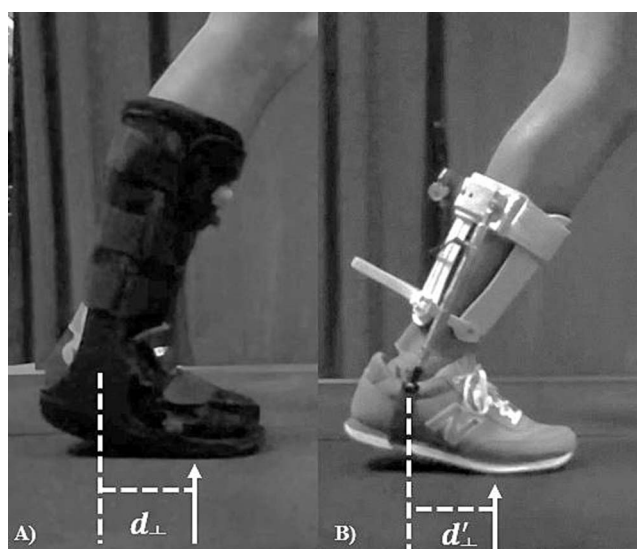


FIGURE 6—Achilles tendon moment arm relative to the ankle joint in the walking boot (A) and DAO (B).

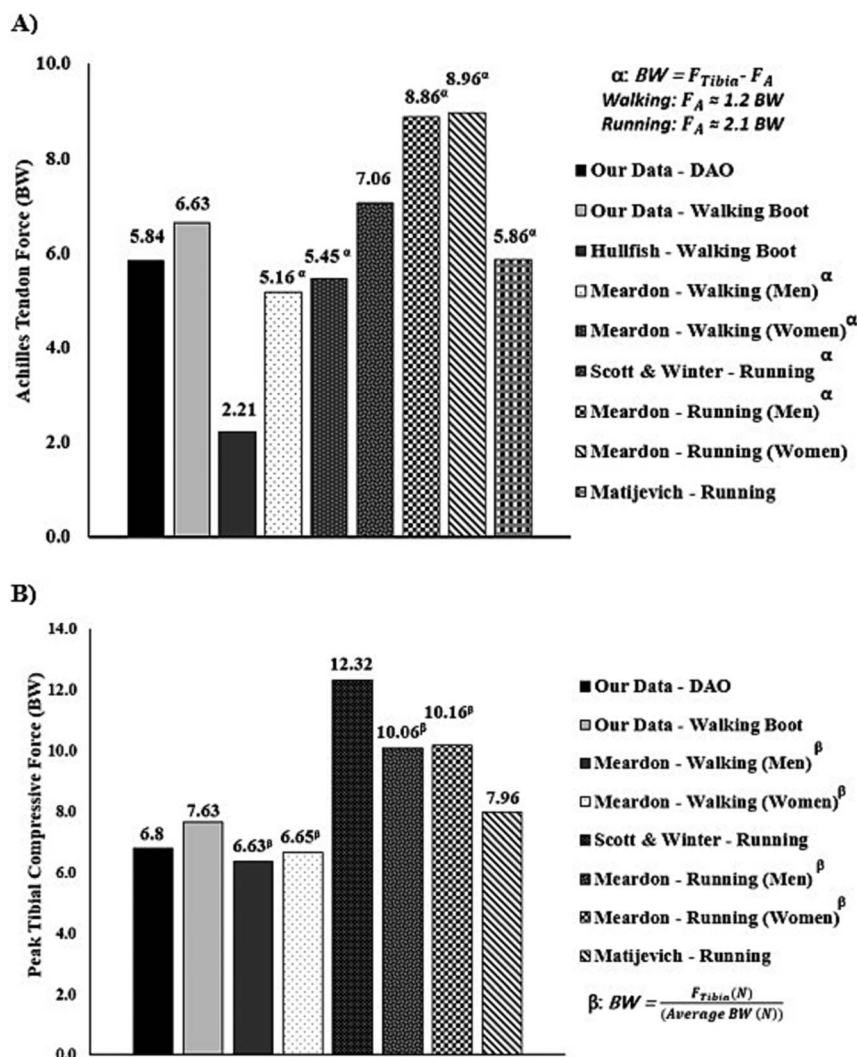


FIGURE 7—Achilles tendon force expressed as BW from the current study compared with previous literature. α : Studies only provided the tibial compressive load. The Achilles tendon force was estimated by assuming the reaction force at the ankle to be similar to the GRF for walking or running, and subtracting the value from the tibial compressive load, provided an estimated value of the (A) Achilles tendon force and (B) peak tibial compressive force expressed as BW from the current study compared with previous literature. β : Studies provided the tibial compressive force in newtons. The tibial compressive force was divided by the participant average body weight to provide an estimate in units of body weight.

force insoles and instrumented treadmill may not provide the exact same GRF values. However, use of force insoles inside the shoe, between the brace and the foot, was important to capture the offloading effect of each brace condition during testing. In this study, treadmill walking was chosen to control participant walking speed during testing, and therefore, current findings may not be generalizable to overground conditions.

CONCLUSIONS

The findings from this study indicated that the DAO moderately reduced tibial compressive force and Achilles tendon force and allowed more sagittal ankle excursion during tread-

mill walking compared with an orthopedic boot. These acute differences between the DAO and a standard-of-care walking boot could have positive benefits on TBSI rehabilitation. Future studies will utilize cadaver specimens to measure and confirm if the DAO is reducing the tibial bone strain. In addition, we plan to investigate the long-term effects of DAO wear on bone health and healing in athletes recovering from TBSI.

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